LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Functional diagram of a basic closed loop flight control system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Functional diagram of a basic closed loop flight control system. Explanation of automatic control and classification of system and explanation of controls.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of functional block diagram components; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define automatic control in the context of flight systems.
(c) List the basic components of a closed loop system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Automatic Control • Closed Loop System Characteristics • Functional Block Diagram Elements • Signal Flow and Feedback Path • System Stability Criteria	L & Ds	PPT, IB	15 min
(b)	• Classification of Control Systems • Single Axis vs Multi-Axis Systems • Control Surface Actuation • Sensor Inputs and Processing • Actuator Response Mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Explanation of System Controls • Pilot Input Integration • Error Signal Generation • Final Control Element Operation • System Loop Closure Verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the error signal is generated in a closed loop system.
(b) Differentiate between open loop and closed loop flight control systems.
(c) Describe the role of the final control element in the AFCS.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between single axis, two axis, and three axis flight control systems for the next lesson (LP 17).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Classification of FCS: - single axis, two axis, and three axis.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min